

# Experiment ‘p\_6\_2\_d\_replace\_with\_larger\_neighbor\_seq’ Results

June 5, 2026

**Experiment outcome:** SUCCESS  
**Bad responses:** 0  
**Responses containing assume :** 0  
**Responses containing expect :** 0  
**Resolution attempts:** 1  
**Hard fails (resolution):** 0  
**Soft fails (resolution):** 0  
**Verification attempts:** 1

## Problem Specification

**Problem name:** p\_6\_2\_d\_replace\_with\_larger\_neighbor\_seq

**Natural language statement:** null

**Method signature:** p\_6\_2\_d\_replace\_with\_larger\_neighbor\_seq(inputs: seq<int>) returns (result: seq<int>)

### Ensures

- `|result| == |inputs|`
- `result[0] == inputs[0]`
- `result[|inputs| - 1] == inputs[|inputs| - 1]`
- forall `i :: 1 <= i < |inputs| - 1 ==> result[i] == if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1] else inputs[i + 1]`

### Requires

- `|inputs| >= 3`

## GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

### Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_d_replace_with_larger_neighbor_seq(inputs: seq<int>) returns (result: seq<int>)`

The method should respect the following contract:

```
requires |inputs| >= 3, ensures |result| == |inputs|, ensures result[0] == inputs[0],
  ↪ ensures result[|inputs| - 1] == inputs[|inputs| - 1], ensures forall i :: 1 <= i <
  ↪ |inputs| - 1 ==> result[i] == if inputs[i - 1] >= inputs[i + 1] then inputs[i -
  ↪ 1] else inputs[i + 1]
```

Produce and show only the Dafny body of this method, including the curly braces that  
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor  
↪ 'expect' anywhere in your code.

### GenAI → Program

**System fingerprint:** null

**ID:** chatcmpl-DnNmR3vIHfplvuYkiRsFwBmRm3jfS

**Model:** gpt-5.5-2026-04-23

**Created at:** 1780661855

**Finish reason:** stop

**Usage:** 434 tokens in, and 735 tokens out

```
{
  var n := |inputs|;
  result := [inputs[0]];
  var i := 1;
  while i < n - 1
    invariant n == |inputs|
    invariant 1 <= i <= n - 1
    invariant |result| == i
    invariant result[0] == inputs[0]
    invariant forall j :: 1 <= j < i ==> result[j] == if inputs[j - 1] >= inputs[j + 1]
      ↪ then inputs[j - 1] else inputs[j + 1]
    decreases n - 1 - i
  {
    var v := if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1] else inputs[i + 1];
    result := result + [v];
    i := i + 1;
  }
  result := result + [inputs[n - 1]];
}
```

```
}
```

## Final Program

```
method p_6_2_d_replace_with_larger_neighbor_seq(inputs: seq<int>) returns (result: seq<
  ↪ int>)
  requires |inputs| >= 3
  ensures |result| == |inputs|
  ensures result[0] == inputs[0]
  ensures result[|inputs| - 1] == inputs[|inputs| - 1]
  ensures forall i :: 1 <= i < |inputs| - 1 ==> result[i] == if inputs[i - 1] >=
    ↪ inputs[i + 1] then inputs[i - 1] else inputs[i + 1]
{
  var n := |inputs|;
  result := [inputs[0]];
  var i := 1;
  while i < n - 1
    invariant n == |inputs|
    invariant 1 <= i <= n - 1
    invariant |result| == i
    invariant result[0] == inputs[0]
    invariant forall j :: 1 <= j < i ==> result[j] == if inputs[j - 1] >= inputs[j + 1]
      ↪ then inputs[j - 1] else inputs[j + 1]
    decreases n - 1 - i
  {
    var v := if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1] else inputs[i + 1];
    result := result + [v];
    i := i + 1;
  }
  result := result + [inputs[n - 1]];
}
```

## Total Token Usage

Input tokens: 434

Output tokens: 735

Reasoning tokens: 512

Sum of 'total tokens': 1169

## Experiment Timings

**Overall Experiment** started at 1780661853698, ended at 1780661893148, lasting 39450ms (39.45 seconds)

**Iteration #1** started at 1780661853716, ended at 1780661893148, lasting 39432ms (39.43 seconds)